EE337 Microprocessors Laboratory

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Problem set **Date: March 26, 2026**

1. [**30 points**] This Project is the first step of an electical engineer to become a firmware/Device driver developer. The main focus of the project is to provide the approach to write a basic device driver on any microcontroller board. The project focuses on establishing serial communication between the 8051-based PT-51 microcontroller and a BMP280 Based Pressure sensor breakout board using the I2C protocol. The primary objective is to develop a software-based I2C code utilizing I2C hardware on 8051 on the PT-51 board and then port it to full fledged application layer functions called Device drivers (Here the programmer does not know what is happening to internal registers). Using this custom driver, the microcontroller will act as the master device to communicate with any generic slave device.

   Phase 1- Week 1 [10 points]

   Write a baremetal code to read the identification/whoami register of the I2C device (BMP280) for PT-51 board.Use LCD for debugging the code and check the intermediate value of registers.

   Approach: Understand I2C- White paper from Texas Instruments

   Read the two wire interface section (from the datasheet) of 8051 microcontroller

   Read The Register map section of BMP280

   Note : Pinmap of the Microcontroller(from AT89C5131A) on the PT-51 is given on the last page of this document

   Link of the complete datasheet(Not Required to read it only for reference)

   Phase 2- Week 2 [15 points]

   Write your own Device driver library to read and write to any arbitrary I2C device memory address.

   The library should contain the following function prototypes.

   - `I2C_Start()`
   - `I2C_Stop()`

- `I2C_Write()`
- `I2C_Read()`
- `I2C_Ack()`
- `I2C_Nack()`

Create a .h file for function prototypes and create .c file for function definitions.

The students can add their own function prototypes for their aid.

Phase 3- Week 3 [5 points]

In this lab session, you will use the Inter-Integrated Circuit (I2C) Driver developed by you to interface a Real-Time Clock (RTC) Module with the 8051 microcontroller. You will use this to set up and read timeStamps at regular intervals from it and display the timeStamp on the onboard LCD Display of PT-51 board. Initialize the RTC and temperature sensor, and perform sequential read operations to fetch the current time registers (Seconds, Minutes, and Hours). The retrieved time data, natively formatted in Binary-Coded Decimal (BCD), will then be processed and accurately displayed on the LCD, demonstrating successful bidirectional master-slave communication.Also print the time and temperature data to PC via UART using serial terminal.

Generic uart driver of understanting

Datasheet of RTC Module:

Use the previously provided `serial.h` library (for UART Communication) as a reference to complete this lab exercise. Make sure that all the DIP switches are OFF (If you are using any GPIO pins on Port1 for I2C).

Note:

1. Use 1200 baud rate for uart.
2. RTC needs 200 ms upon initial power-up.
3. DS1307 works with an I2C speed of 100 kHz.
4. Use 10 kΩ pull-up resistors across SDA and SCL lines of I2C bus for reliable operation.

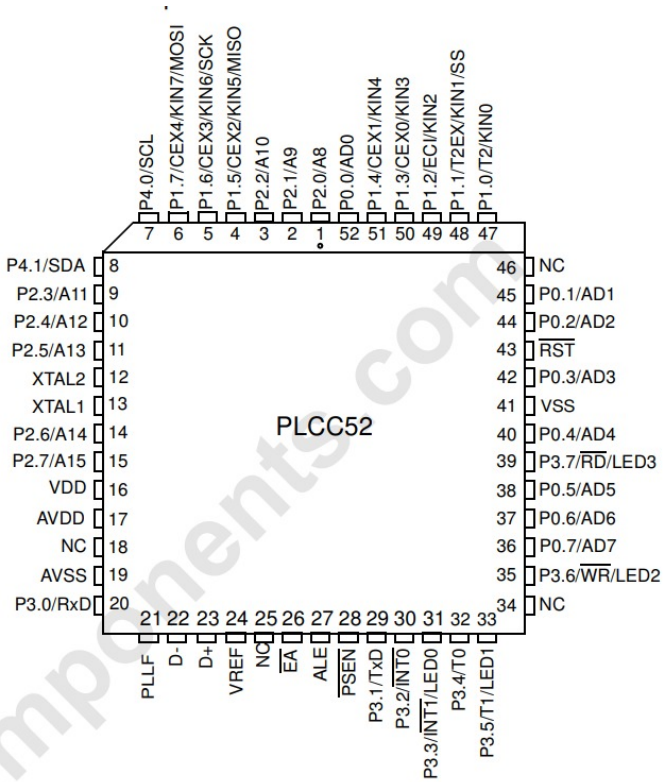

Figure 1: Pinmap of 8051 microcontroller on PT-51 board